

X3 / LIVE READINESS

THE ROAD TO MAINNET

**20/100 evidence readiness. 0/29 tasks completed.
Release decision: NO-GO.**

2026-09-05T19:53:53.366172+00:00

Historical audit credit remains eligible on unchanged source.

A score of 100 never grants launch approval. Closing the tracked findings changes the automated decision to NOT ASSESSED; independent launch-gate approval is still required.

20 points for each implemented/wired/tested/executed/reproducible criterion; weighted subsystem mean; cap 20 while a Critical finding remains open.

Source: 475338a94dcc01f7ad3a3617ccb78fef42d8f85d78aca18a0ae21333d759151e

Named local operator attestations, not authenticated independent security certification. Local writers can alter state and evidence; hashes detect accidental tampering, not a hostile store owner.

Figure 1 · Readiness evidence



Figure 2 · Readiness by subsystem

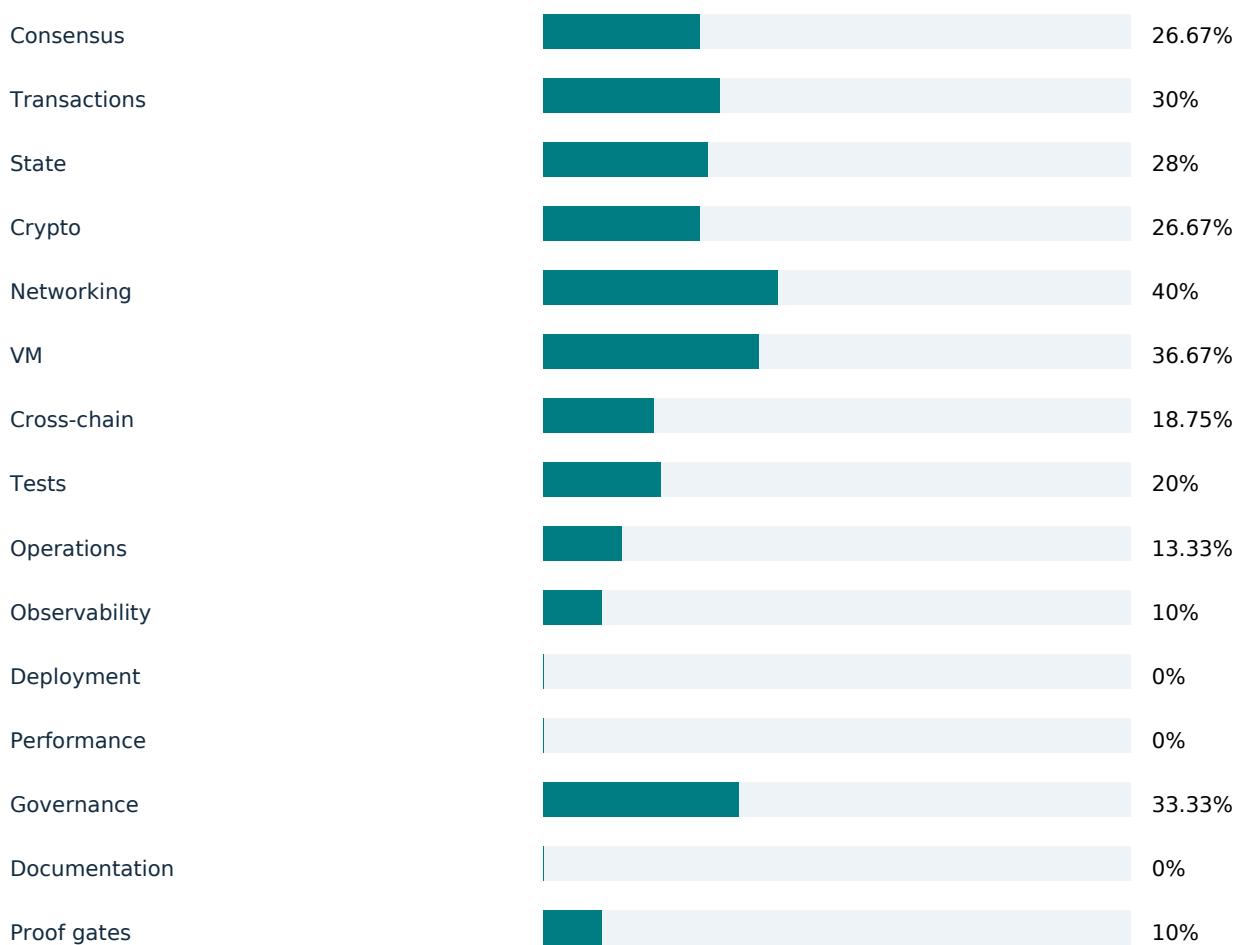


Figure 3 · Open findings



Figure 4 · Task status

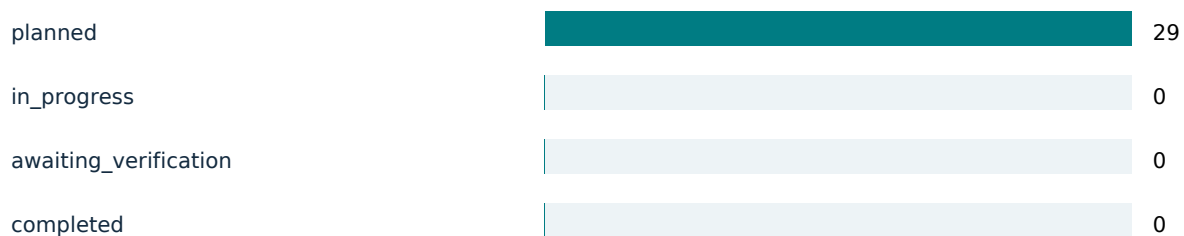


Figure 5 · Verification status



Tasks and finding closure

ID / severity	Task	Current status	Acceptance requirement
FIX-C01 / Critical	External header validation accepts unproved claims	planned	Arbitrary hashes, duplicate signers, self-selected validator sets, future heights and invalid membership proofs are rejected without writes; canonical external fixtures pass.
FIX-C02 / Critical	Unsigned finality anchors make certificate checks circular	planned	Forged and conflicting anchors cannot enter pool or dispatch; independently verified justification anchors the exact finalized block; late legitimate anchors remain recoverable.
FIX-C03 / Critical	Unsigned rollback receipts trust attacker-selected prior state	planned	Unrelated caller/diff and stale pre-state are rejected; injected rollback failure leaves no partial state; reverse-order replay restores exact pre-root.
FIX-H01 / High	Production proof-router alternatives accept arbitrary bytes	planned	All three isolated rejection tests pass; actual signed finalized fixtures pass through the runtime gateway, not only the helper.
FIX-H02 / High	EVM receipt verifier lacks trusted roots and misbinds inclusion	planned	Cross-check independent execution-client proof vectors, tampered receipt values and false heads; require both authentic root and exact inclusion.
FIX-H03 / High	Bitcoin verification uses asserted confirmations and incomplete vault approval	planned	Reject invented tips, wrong outputs, wrong recipient/value, repeated approvals and forged signatures; restart and reorg tests preserve accounting.
FIX-H04 / High	Public misbehavior report has no evidence and does not slash currency	planned	Invalid/replayed evidence cannot slash; total reserved balances and treasury/burn conservation reconcile after each valid offence; removed keys cannot author/vote.
FIX-H05 / High	Validator rotation ignores requested activation delay at session boundary	planned	A change delayed over several sessions is not activated early; restart and session transition preserve identical authority sets on all nodes.
FIX-H06 / High	Flash-finality opt-in disables GRANDPA without a proven replacement	planned	Wrong keys/sets cannot form certificates; missing keys fail startup; partition, equivocation and recovery tests establish one finalized history.
FIX-H07 / High	Default mini-EVM executes against disposable synthetic state	planned	A contract SSTORE persists across blocks/restart, a second user cannot impersonate the first, out-of-gas fully reverts and fees reconcile.
FIX-H08 / High	Ethereum submission RPC bypasses the transaction-pool lifecycle	planned	Submit a signed transfer over RPC, observe pool admission, block inclusion, finality and persisted balance change on another node; reject replay/wrong chain ID.

ID / severity	Task	Current status	Acceptance requirement
FIX-H09 / High	Relayer manual extrinsic encoding and signing do not match runtime	planned	Exact production runtime decodes, authenticates, includes and finalizes generated deposit/release transactions; no fallback signer or zero-chain context.
FIX-H10 / High	Relayer checkpoints can skip failed events and are not durable	planned	Fail submission for an early event, process later events, restart and recover the failed event exactly once; bound RPC scans and memory.
FIX-H11 / High	Gateway binary never starts its REST/GraphQL implementation	planned	Start the built binary against disposable services, serve actual API/GraphQL data, fail readiness when DB is down, and recover after restart.
FIX-H12 / High	SDK EVM and SVM encoding contains production placeholders	planned	Match independent reference vectors for EVM selectors/addresses and multi-account SVM messages, then finalize one transaction per advertised protocol.
FIX-H13 / High	CI make recipes swallow failures and reference missing targets	planned	Cargo failure produces nonzero make/CI exit; every workflow command resolves; mainnet-required aggregate cannot pass with skipped/missing jobs.
FIX-H14 / High	Release gate accepts stale build artifacts without source binding	planned	Mutating source or substituting a binary/spec invalidates evidence and blocks release; two builders reproduce the same WASM.
FIX-H15 / High	Dependency audit pass masks unresolved advisory matches	planned	No reachable unmitigated critical/high advisories; exceptions have owners, expiry, exact dependency paths and regression evidence.
FIX-H16 / High	Supply invariant finalization work is unbounded by asset count	planned	Worst-case asset count fits declared block budget; adversarial asset creation cannot make next-block mandatory work unbounded.
FIX-H17 / High	EVM external verifier is blocked in production and bypassable by mode	planned	Real proof accepted, zero/duplicate/forged signatures rejected; funded deployment cannot enable structural bypass; owner rotation preserves trust policy.
FIX-M01 / Medium	Readiness documentation contradicts executable evidence	planned	Any failed core proof lowers its status; stale commit evidence is rejected; readiness prose and matrix derive from the same data.
FIX-M02 / Medium	Finality and multi-node proof tests are ignored	planned	At least four validators agree on finalized hash/root through restart and one-node failure; ignored critical tests count as release failure.
FIX-M03 / Medium	Gateway health and event spines do not prove dependencies	planned	Stop database/event sink and see readiness fail; replay missed events exactly once after recovery.
FIX-M04 / Medium	Restore validates archive shape, not trusted checksum or chain state	planned	Tampered archive fails before extraction; restored node catches up and matches independently recorded finalized state.

ID / severity	Task	Current status	Acceptance requirement
FIX-M05 / Medium	Python DSL emitter tests fail under the current proof contract	planned	All emitter tests pass with independently validated proof fixtures and reject empty/tampered bundles; one local end-to-end emission is consumed by the real runtime.
FIX-M06 / Medium	Desktop defaults select an unregistered chain	planned	Default startup obtains the intended adapter, missing network fails clearly, and wrong genesis cannot be signed against.
FIX-M07 / Medium	Optimizer throughput is not network TPS evidence	planned	Publish reproducible sustained finalized TPS, rejection rate, p50/p95/p99 finality latency and hardware/profile/commit details; no synthetic metric substitution.
FIX-L01 / Low	Fresh-machine helper does not verify the required environment	planned	Clean image builds all intended targets and runs real smoke tests with explicit offline/cache prerequisites.
FIX-H18 / High	Workspace verification fails in WASM runtime build	planned	From the frozen source state, all required workspace checks and release/testnet feature builds exit zero and produce source-bound artifacts.

Verification runs

Check	Command	Result	Evidence
rpc-unit	cargo test --locked --manifest-path audit-artif acts/mainnet-readiness/20 26-09-05-6a24d8cf-audit/a udit-harness/rpc/Cargo.to ml	passed: Fresh execution and matching evidence hashes	7795cf968e0d4a129367ac81f926c246 / exit 0
dsl-tests	/usr/bin/python3 -m pytest -q x3-lang/tests/test_parse r.py x3-lang/tests/test_typ echecker.py x3-lang/tests/ test_emitter.py x3-lang/te sts/test_simulator.py	not_run: No recorded execution	Not run

Feature evidence

Feature	Score / status	Criterion provenance
FT01 Aura block production	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT02 GRANDPA finality and fork choice	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT03 Validator rotation	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT04 Validator bonded stake and slashing	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT05 Flash finality	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT06 Genesis construction and live-seed validation	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT07 Signed native transaction path	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT08 Nonce, genesis, era replay checks	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT09 Transaction pool limits and ordering	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT10 Fee charging and refunds	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT11 Ethereum raw transaction RPC	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT12 Canonical FRAME storage	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT13 Storage upgrade migrations	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT14 Supply conservation ledger	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified

Feature	Score / status	Criterion provenance
FT15 Snapshot backup and restoration	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT16 Atomic rollback state provenance	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT17 Signature primitives and key custody	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT18 Dependency advisory handling	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT19 P2P networking and sync	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT20 Bootstrap helper configuration	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT21 PoH import verification	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT22 RPC limiter algorithm (narrow scope)	100% / VERIFIED	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: historical audit, unchanged source; executed: historical audit, unchanged source; reproducible: historical audit, unchanged source
FT23 WASM mini-EVM persistent execution	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT24 SVM execution and account context	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT25 X3 VM runtime	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT26 X3 Python parser and typechecker	100% / VERIFIED	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: historical audit, unchanged source; executed: historical audit, unchanged source; reproducible: historical audit, unchanged source
FT27 X3 emitter and execution pipeline	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT28 Rust compiler tracks	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified

Feature	Score / status	Criterion provenance
FT29 Internal cross-VM representation router	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT30 Atomic bundle orchestration / Atomic Lock	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT31 Settlement timeout and refund engine	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT32 External header finality oracle	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT33 Production EVM receipt proof route	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT34 Solana finalized proof route	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT35 Validator quorum proof route	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT36 Bitcoin SPV route	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT37 Bitcoin vault approvals	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT38 Relay binary delivery	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT39 Relay typed submission library	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT40 EVM HTLC contract	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT41 SVM HTLC contract	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT42 External EVM proof verifier contract	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT43 Solver marketplace and intent routing	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified

Feature	Score / status	Criterion provenance
FT44 Validator attestation / proof ledger	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT45 REST/GraphQL gateway	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT46 SQL indexer and migrations	20% / PARTIAL	implemented: historical audit, unchanged source; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT47 Metrics / chain health monitoring	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT48 Security and accounting event consumers	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT49 DEX, Forge and LP locker	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT50 Sentinel and economic halt	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT51 Council / treasury / upgrades	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT52 Wallet / biometric / recovery pallet	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT53 TypeScript SDK encoding	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT54 Desktop / Tauri OS network integration	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT55 GPU / swarm orchestration	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT56 CI test quality and coverage	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT57 Workspace build / release reproducibility	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT58 Fresh-machine bootstrap	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified

Feature	Score / status	Criterion provenance
FT59 Deployment isolation and validator services	40% / PARTIAL	implemented: historical audit, unchanged source; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT60 Sustained finalized TPS evidence	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT61 Mainnet / ProofGate enforcement	20% / PARTIAL	implemented: unverified; wired: historical audit, unchanged source; tested: unverified; executed: unverified; reproducible: unverified
FT62 Documentation accuracy / scoreboard	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT63 Public testnet recovery drill evidence	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified
FT64 Independent launch approval evidence	0% / UNVERIFIED	implemented: unverified; wired: unverified; tested: unverified; executed: unverified; reproducible: unverified

Closure and criterion reviews

No records yet.